ROHAN BAHORDER

Cybersecurity and AI/ML | Post-Quantum Cryptography Research

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PROFESSIONAL SUMMARY

Computer Science graduate from BRAC University with research and project experience in cybersecurity and machine learning, specializing in post-quantum cryptography and privacy-preserving AI.

- **Designed and implemented** hybrid post-quantum encryption (ML-KEM, ML-DSA, AES-256) to secure federated learning across 20 clients as an undergraduate thesis.
- **Built** end-to-end machine learning pipelines, including an original three-source football dataset with models about 21% better than baseline.
- **Trained** deep generative models (LSTM autoencoder, VAE, Transformer with RLHF-style tuning) in PyTorch and benchmarked them against baselines.
- **Solved** real research obstacles: redesigned the thesis test method when lab restrictions blocked the original plan and kept the research on track.
- **Approached** every university research and project task as professional work, with clear goals, careful testing and complete delivery.
- **Combines** security and machine learning skills, with a strong interest in privacy-preserving ML and AI-driven security.

CORE COMPETENCIES

Post-Quantum Cryptography | Applied Cryptography | Secure Communication Protocols | Federated Learning | Machine Learning and Deep Learning | Data Analysis and Feature Engineering | Network Programming | Experimental Benchmarking | Technical Research and Reporting | Problem Solving | Team Collaboration

EDUCATION

B.Sc. in Computer Science, BRAC University, Dhaka | 2022 to 2026

Higher Secondary Certificate (HSC), Al-Amin Academy School and College, Chandpur | 2020 | GPA 5.00 / 5.00

RESEARCH EXPERIENCE

Post-Quantum Hybrid Cryptography for Federated Learning | Undergraduate Thesis, BRAC University | Python, Flower, ML-KEM, ML-DSA, AES-256

- Designed three hybrid encryption schemes at rising NIST security levels, combining ML-KEM (512, 768, 1024) key exchange, ML-DSA (44, 65, 87) signatures and AES-256 encryption.
- Integrated the schemes into a Flower federated learning pipeline to protect model updates exchanged between 20 clients and a central server over a dedicated secure socket channel.
- Hardened the protocol with HKDF-SHA-256 key derivation, an acknowledgement handshake and structured error logging, and resolved inconsistencies across scheme versions.
- Built a multi-machine lab testbed and a nanosecond-precision timing framework to measure key generation, encapsulation, signing and encryption overhead per client, per round.
- Designed a per-client ratio analysis across identically configured machines to compare security levels fairly under lab hardware constraints.

Predicting Penalties Won in Football | Independent Research | Python, pandas, scikit-learn, XGBoost

- Collected, cleaned and merged team and player statistics from FBref, FotMob and FPL across 5 Premier League seasons into an original dataset of 100 team-seasons and about 550 players per season.
- Engineered features for build-up play, midfield creation and forward threat; benchmarked Linear Regression and XGBoost with 5-fold cross-validation.
- Achieved RMSE 1.908 with the best model (45 of 100 predictions within one penalty), about 21% better than baseline.
- Traced the largest prediction errors to an individual penalty-specialist effect, a finding not captured by team-level features.

PROJECTS

Multi-Genre Music Generation with Deep Generative Models | Python, PyTorch | github.com/rohanMB357/neural-network-project

- Built a MIDI pipeline that parses and tokenizes music across 5 genres (classical, jazz, rock, pop, electronic).
- Trained and compared an LSTM autoencoder, a variational autoencoder and an autoregressive Transformer decoder against random and Markov chain baselines.
- Fine-tuned the Transformer with RLHF-style policy gradient using human survey scores as rewards, and exported generated samples to MIDI.

Network Design and Socket Programming | Python, TCP/UDP, Cisco Packet Tracer | github.com/rohanMB357/cse421

- Implemented TCP and UDP client-server applications in Python, handling connection setup, message exchange and error cases.
- Designed and simulated a multi-router network topology in Cisco Packet Tracer with routing, switching and serial links.

Water Pollution Classification | Python, scikit-learn, pandas | github.com/rohanMB357/Machine-learning-

- Built a supervised classifier for water quality and pollution levels, covering data cleaning, feature selection, training and evaluation.
- Reviewed related research and wrote a paper summary to guide model and feature choices.

EDU Flow: Course Management System | PHP, MySQL, JavaScript | github.com/rohanMB357/edu-flow

- Built a role-based web application for students and faculty with separate access, login and authorization flows.
- Implemented course enrollment, assignment submission, quizzes, material sharing, messaging and result tracking on a MySQL database.

Car Repair Management System | MongoDB, Express, React, Node.js

- Built a full-stack MERN application with service booking, customer management and end-to-end CRUD workflows.

TECHNICAL SKILLS

Cryptography and Security: ML-KEM (Kyber), ML-DSA (Dilithium), AES-256, HKDF-SHA-256, hybrid key exchange, digital signatures, secure socket communication, key management

Machine Learning and AI: PyTorch, scikit-learn, XGBoost, Flower (federated learning); LSTMs, VAEs, Transformers, regression, classification, cross-validation

Data: pandas, NumPy, Matplotlib, Jupyter, data cleaning, feature engineering, multi-source data integration

Networking: TCP/IP, UDP, socket programming, routing and switching, Cisco Packet Tracer

Programming: Python, C++, JavaScript, PHP, SQL

Web and Tools: React, Node.js, Express, MongoDB, MySQL, Git, GitHub, Linux

RESEARCH INTERESTS

Post-quantum cryptography | Privacy-preserving and federated machine learning | AI for cybersecurity | Secure system design | Adversarial machine learning

AWARDS AND LANGUAGES

- Participant, Bangladesh Mathematics Olympiad
- Languages: Bengali (native), English (professional proficiency)